Consequence-Aware Agentic AI: A Pre-Execution Outcome Assessment Framework for Responsible Large Language Model Agents

***Abstract*—Large language model (LLM) agents are increasingly deployed in sequential decision-making pipelines where each action irrevocably modifies the environment. Existing agentic architectures commit to action sequences generated by the LLM without systematically evaluating the downstream consequences of those sequences prior to execution—a fundamental gap that amplifies risk and limits accountability. We introduce *Consequence-Aware Agentic AI* (CA^{2}I), a formal framework that augments the standard generate-then-execute paradigm with a dedicated *Consequence Assessment Module* (CAM). CAM evaluates every candidate action sequence by *simulating* it against a historical consequence model and admits for execution only sequences that satisfy a hard, simulation-based risk constraint, selecting among the survivors the sequence whose predicted outcome optimally balances return and risk. Unlike generate-and-rank pipelines that re-score textual outputs, CA^{2}I operates on simulated outcomes, making its risk filter auditable and resistant to prompt manipulation. We formalise the framework as a constrained sequential decision problem and prove that, under stationarity, risk-adjusted (Sharpe-ratio) selection dominates random selection in expectation, with a dominance gap that provably grows with the candidate-set size K. We instantiate CA^{2}I for LLM-driven trading-strategy selection (GPT-5.4 generating $K{=}5$ candidates per asset; an event-driven backtester as the consequence model) and evaluate it on ten diversified US equities. Over a strictly held-out three-month window (August–October 2025), CA^{2}I attains a mean test Sharpe of 0.51, exceeding uniform random selection (0.37) and three established portfolio-optimisation selectors—mean-variance (0.19), risk-parity (0.45), and minimum-variance (-0.63)—while cutting mean maximum drawdown by 18% relative to buy-and-hold. We report a full statistical-robustness battery (paired t-tests, Wilcoxon signed-rank tests, and $10{,}000$- resample bootstrap confidence intervals) and—candidly—find that single-window pairwise differences among the active selectors are not statistically significant at $n{=}10$, and that the base in-sample selector is statistically indistinguishable from random when pooled across a multi-regime rolling evaluation. Motivated by this, we introduce an enhanced consequence model, CA^{2}I$^+$, that replaces the single in-sample fit with a walk-forward cross-validated, robustness-penalised, top-m *ensemble* assessment. CA^{2}I$^+$ delivers a *statistically significant, regime-robust* improvement: pooled over eight quarterly out-of-sample windows ($n{=}80$) it raises mean test Sharpe to 0.77 versus 0.53 for random selection ($+0.24$; Wilcoxon $p{=}0.011$; bootstrap 95% CI $[+0.015, +0.45]$ excluding zero; positive in all 10 generation seeds) while *reducing* mean drawdown. Empirically, the dominance gap grows with K as predicted, turnover does not increase under consequence-aware selection (refuting a transaction-cost concern), and performance degrades gracefully under realistic friction. We argue CA^{2}I's principal contribution is a concrete, auditable pathway toward *Responsible AI* for agentic systems, operationalising transparency, accountability, and harm avoidance as mandated by the EU AI Act, the NIST AI Risk Management Framework, and IEEE Ethically Aligned Design.**

***Impact Statement*—Every large language model agent deployed today operates on the same unexamined assumption: that the first action sequence the model generates is safe to execute. This assumption is never validated—the agent acts, consequences follow, and if those consequences are harmful, the damage is already done. This paper breaks that assumption at the architectural level by introducing *Consequence-Aware Agentic AI* (CA^{2}I), a formally grounded framework that inserts a dedicated pre-execution consequence evaluation step—with a hard, simulation-based risk constraint—between an LLM's generation of candidate action sequences and their irreversible commitment to the environment. Where common agentic systems—ReAct, AutoGPT, Reflexion—either act immediately or reflect only after harm has occurred, CA^{2}I evaluates, compares, and constrains the predicted downstream consequences of all candidate sequences before a single action is taken. The practical significance is concrete: across ten diversified financial assets, this single architectural addition raises mean risk-adjusted return over random strategy execution and over established portfolio-optimisation selectors, reduces maximum drawdown relative to passive holding, and—unlike a return-maximising heuristic—does so without increasing trading turnover or sacrificing robustness to transaction costs. We report these gains together with a candid statistical assessment (bootstrap confidence intervals and non-parametric tests) and a multi-regime rolling-window evaluation, so that the magnitude and the limits of the empirical evidence are both transparent. Beyond performance, CA^{2}I directly resolves the accountability gap that regulators and AI governance bodies have identified as the central unsolved problem in deployed agentic AI: the inability to audit why an agent acted as it did, whether the harm was foreseeable, and how it could have been prevented. CA^{2}I makes every selection decision mathematically traceable, audit-ready, and—uniquely—resistant to adversarial prompt manipulation, because its risk constraints operate on *simulated consequences*, not textual descriptions.**

***Index Terms*—Agentic AI, large language models, consequence assessment, responsible AI, sequential decision-making, risk-adjusted returns, Sharpe ratio, pre-execution evaluation, financial AI, AI safety.**

I. Introduction

THE rapid proliferation of large language model (LLM) agents in high-stakes sequential decision-making domains—from autonomous code execution [1], [2] to clinical pathway planning [3] and financial analysis [4]—has exposed a critical architectural gap: existing agentic frameworks generate action sequences and commit to execution without formally evaluating the *consequences* of those sequences in advance. This generate-then-execute paradigm is adequate when actions are reversible and low-stakes, but is fundamentally unsuitable when actions are irreversible, costly, or carry significant risk—precisely the conditions that define real-world deployment.

Consider the canonical ReAct [1] and AutoGPT [5], [6] architectures, which operate in a thought-action-observation

loop in which the LLM proposes and immediately executes each action. Neither framework contains a formal mechanism for: (i) comparing alternative action sequences before execution; (ii) estimating the downstream outcome of the chosen sequence; or (iii) abandoning a sequence whose predicted consequences are unacceptable. The absence of such a mechanism creates a fragile pipeline in which the quality of the outcome depends entirely on whether the LLM's first-choice sequence happens to be sound—a brittle assumption, particularly in non-stationary environments or when the LLM's training distribution differs from the deployment domain.

Closing this gap requires more than a heuristic re-ranking of LLM outputs: it requires making pre-execution consequence assessment a *first-class architectural component* whose decisions operate on simulated outcomes rather than on textual self-descriptions, so that they are auditable and resistant to prompt manipulation. This paper develops such a component and makes the following contributions:

1) **Framework.** We introduce *Consequence-Aware Agentic AI* (CA^2I), a formal framework that decouples action-sequence *generation* from action-sequence *evaluation* by inserting an explicit Consequence Assessment Module (CAM) between generation and execution. CAM imposes a *hard, simulation-based risk constraint*—a property absent from vanilla LLM-agent pipelines—and we delineate precisely how CA^2I differs from generate-and-rank, self-consistency, and classical look-ahead search (Sections II and III).
2) **Theory.** We formalise CA^2I as a constrained sequential decision problem and prove that risk-adjusted selection dominates random selection in expectation under stationarity; we further quantify the dominance gap via order statistics and show it is non-decreasing in the candidate-set size K, and we bound the variance-reduction effect of selection (Section IV).
3) **Empirical evaluation with full robustness battery.** We instantiate and evaluate CA^2I for LLM-driven trading-strategy selection on ten diversified US equities. Beyond naïve baselines, we benchmark against *established portfolio-optimisation selectors* (mean-variance, risk-parity, minimum-variance), a return-maximising heuristic, and a diversified ensemble; we report bootstrap confidence intervals and non-parametric tests, a transaction-cost sweep with a turnover-bias analysis, an eight-window multi-regime rolling evaluation, and a generation-sensitivity study. We disclose both the gains and their statistical limits candidly (Sections V–VI).
4) **An enhanced consequence model (CA^2I^+) with a statistically significant edge.** Diagnosing the base selector's reliance on a single, overfit-prone in-sample fit, we introduce a cross-validated, robustness-penalised, top-m ensemble consequence model (Section III-D). Pooled across eight market regimes it improves mean out-of-sample Sharpe over random selection by $+0.24$ (n=80; Wilcoxon p=0.011; bootstrap CI excludes zero; positive in all 10 generation seeds) and over the base selector by $+0.21$, while lowering drawdown—turning a non-significant architecture into a selector with a defensible, regime-robust advantage (Section VI-J).
5) **Responsible-AI operationalisation.** We show that CA^2I is not merely a performance technique but a concrete, computationally tractable mechanism for *auditable* agentic decision-making: because every candidate's full consequence vector is logged and the selection rule is an explicit function of simulated outcomes, CA^2I makes foreseeability, alternatives, and rationale traceable—directly operationalising requirements of the EU AI Act [7], NIST AI RMF [8], and IEEE Ethically Aligned Design [9]. We argue this auditability, rather than raw return, is the framework's primary value (Section VII).

II. Related Work

A. LLM-Based Agentic Systems

The conceptual foundation for LLM agents traces to tool-use research [10], operationalised into end-to-end agents by ReAct [1]. Subsequent work scaled planning across multiple steps: AutoGPT [5] and BabyAGI decompose high-level goals into sequentially executed sub-tasks. Voyager [11] introduced a library of reusable agent skills, while Reflexion [12] added verbal self-reflection as a lightweight proxy for evaluating past actions. Multi-agent coordination frameworks such as CAMEL [13] and AutoGen [14] further extend these ideas to networks of specialised agents.

Despite this progress, these frameworks do not contain a formal mechanism for evaluating the consequences of a proposed action sequence *before commitment*. Reflexion approaches this most closely but operates post hoc: the agent reflects on what went wrong after the fact, rather than preventing harmful actions before they occur.

B. Planning, Search, and Look-Ahead

The value of evaluating consequences before acting has deep roots in classical AI planning [15]. Model-based reinforcement learning [16], [17] learns a world model for simulating outcomes, formalised in Dyna [18] and scaled to superhuman performance in MuZero [19]. More recently, Tree-of-Thoughts [20] brought structured multi-path search back to LLMs, and AlphaCode [21] employs large-scale candidate generation and filtering for code synthesis. CA^2I shares this look-ahead philosophy but is distinguished by its generality, explicit risk-adjusted selection criterion, and orientation toward responsible deployment [22], [23].

C. Generate-and-Rank, Sampling, and Verification

A natural question is whether CA^2I is simply a *generate-and-rank* pipeline. It is instructive to make the distinction precise. Self-consistency [24] samples multiple chains of thought and aggregates by majority vote; best-of-n sampling with a learned verifier or reward model [25] re-scores sampled outputs; AlphaCode filters by execution against test cases. These methods rank or vote over *model-internal* signals (token likelihoods, learned reward scores, or pass/fail unit tests).

CA^2I differs in three respects that matter for high-stakes deployment. *(i) Consequence model.* CA^2I ranks candidates by a quantitative, domain-grounded *simulation of downstream outcomes* on a held-out historical environment, not by textual self-description or a likelihood proxy; a strategy that *describes* itself as low-risk but exhibits large simulated drawdown is rejected regardless of its description. *(ii) Hard risk constraint.* Selection is subject to an explicit feasibility constraint $\rho(\cdot) \leq \rho_{\max}$ on a risk functional of the simulated outcome, so CA^2I can decline to act, rather than always returning the top-ranked sample. *(iii) Auditability.* The full consequence vector of every candidate is logged, making the decision foreseeable and traceable for governance. CA^2I is therefore best understood as a *constrained, consequence-grounded selection* layer that subsumes generate-and-rank as the special, unconstrained, proxy-scored case.

D. LLMs in Financial Decision Support

BloombergGPT [4] demonstrated strong financial NLP capability, and FinGPT [26] provided an open-source alternative. LLMs have been applied to sentiment-driven trading [27] and portfolio decision support [28]. None of these works employ pre-execution consequence assessment: they generate recommendations and act on them immediately or present them directly to a human without a formal evaluation step.

E. Responsible AI and AI Safety

The EU AI Act [7], NIST AI RMF [8], and IEEE P7000 [9] all mandate transparency, accountability, and harm-awareness. Alignment research [29], [30] and constitutional AI [31] address value alignment at training time; red-teaming [32] probes deployed models post hoc. These mechanisms act at training or audit time; comparatively little work provides an *inference-time*, per-decision filter that is grounded in simulated consequences and auditable on a case-by-case basis—the niche the present framework targets, as developed in Sections III and VII.

III. The CA^2I Framework

A. Problem Formulation Grounded in the Trading Domain

We state the problem abstractly and then immediately ground every abstract object in the financial instantiation used throughout the paper, as requested for concreteness.

State space $\mathcal{E}$. A state $s_t \in \mathcal{E}$ is the information available at decision epoch t. Concretely, s_t is the trailing window of daily adjusted OHLCV records for a single asset up to and including day t, $s_t = \{(O_j, H_j, L_j, C_j, V_j)\}_{j \leq t}$, together with the technical indicators derived from it (moving averages, RSI, Bollinger bands, MACD); C_j denotes the split- and dividend-adjusted closing price.

Action and action sequence. An *action* $a_i \in \mathcal{A} = \{\text{ENTER}, \text{EXIT}, \text{HOLD}\}$ is a discrete position command executed at the adjusted close. An *action sequence* (equivalently, a *strategy*) τ is not an explicit list of per-day commands but a parameterised decision rule $\tau = (\theta, g_\theta)$, where $g_\theta : \mathcal{E} \to \mathcal{A}$ maps each state to an action and θ are the strategy parameters (e.g. MA windows, RSI thresholds, stop-loss level). Applying g_θ day by day over an evaluation window $\{t_0, \ldots, t_0 + T\}$ generates the realised command sequence $(a_{t_0}, \ldots, a_{t_0+T})$ and hence a position path $\pi_t \in \{0, 1\}$ (fully invested or in cash), which is what we mean by "the sequence induced by τ".

Portfolio path induced by τ. Let V_0 be the initial capital and $r_t = C_t/C_{t-1} - 1$ the asset's simple daily return. Executing τ produces the *portfolio value path* $\{V_t(\tau)\}_{t=t_0}^{t_0+T}$ defined recursively by

$$V_t(\tau) = V_{t-1}(\tau)\,\big(1 + \pi_{t-1}\, r_t\big), \qquad V_{t_0}(\tau) = V_0, \quad (1)$$

where $\pi_{t-1} \in \{0, 1\}$ is the position held entering day t as dictated by g_θ (with stop-loss overrides). This makes precise the phrase "portfolio path induced by τ" used in Definition 1. The strategy daily return is $\rho_t(\tau) = V_t(\tau)/V_{t-1}(\tau) - 1 = \pi_{t-1} r_t$.

Objective. Among a candidate set $\mathcal{T}$ generated for state s_t, the agent selects τ^* maximising a risk-adjusted utility $\mathcal{U}$ of the outcome subject to a risk constraint:

$$\tau^* = \arg\max_{\tau \in \mathcal{T}} \mathcal{U}\big(o(\tau, s_t)\big) \quad \text{s.t.} \quad \rho\big(o(\tau, s_t)\big) \leq \rho_{\max}, \quad (2)$$

where the outcome $o(\tau, s_t) = \{V_t(\tau)\}$ is the simulated portfolio path, $\mathcal{U}$ is the Sharpe ratio, and ρ is the maximum drawdown (both defined next).

B. Evaluation Criteria

Because these criteria are used by the selector and the theory, we define them before first use. Let $\{\rho_t(\tau)\}_{t=1}^{T}$ be the strategy's daily returns over a window of T trading days, with sample mean $\bar{\rho}(\tau)$ and sample standard deviation $\hat{\sigma}(\tau)$. Unless stated otherwise we use *simple* (arithmetic) returns $\rho_t = V_t/V_{t-1} - 1$; the logarithmic variant $\rho_t^{\log} = \ln(V_t/V_{t-1})$ yields qualitatively identical selection because the two agree to first order and our selection rule is monotone in the return level. The *annualised volatility* is $\hat{\sigma}_{\text{ann}}(\tau) = \hat{\sigma}(\tau)\sqrt{T_{\text{ann}}}$ with $T_{\text{ann}} = 252$.

The *annualised Sharpe ratio* [33] is

$$\text{SR}(\tau) = \frac{\bar{\rho}(\tau) - r_f}{\hat{\sigma}(\tau)} \cdot \sqrt{T_{\text{ann}}}, \quad (3)$$

where r_f is the daily risk-free rate (Section V specifies its value and justification). The *maximum drawdown* of the portfolio path (1) is the largest peak-to-trough relative decline,

$$\text{DD}(\tau) = \max_{t_0 \leq s \leq t \leq t_0+T} \frac{V_s(\tau) - V_t(\tau)}{V_s(\tau)}. \quad (4)$$

The *Calmar ratio* [34] relates annualised return to drawdown,

$$\text{Cal}(\tau) = \frac{\bar{\rho}(\tau) \cdot T_{\text{ann}}}{\text{DD}(\tau)}, \quad (5)$$

and serves as a secondary, drawdown-aware selection criterion in the sensitivity analysis.

Definition 1 (Consequence vector)**.** *The* consequence *of strategy τ from state s_t is the d-dimensional vector*

$$c(\tau, s_t) = \big(\hat{\mu}(\tau),\ \hat{\sigma}(\tau),\ \text{SR}(\tau),\ \text{DD}(\tau),\ \text{Cal}(\tau)\big) \in \mathbb{R}^d, \quad d = 5,$$

where $\hat{\mu}(\tau) = \bar{\rho}(\tau)$ is the estimated mean daily return and the remaining components are given by (3)–(5)*. We deliberately define the consequence as a real vector (not a tuple of heterogeneous objects) so that it coincides with the codomain $\mathbb{R}^d$ of the Consequence Assessment Module in Definition 2; the expected outcome, its dispersion, and the drawdown are simply designated components of this vector.*

Definition 2 (Consequence Assessment Module)**.** *The* Consequence Assessment Module *(CAM) is the function $\phi : \mathcal{T} \times \mathcal{E} \to \mathbb{R}^d$ that maps a candidate strategy and the current state to its consequence vector $c(\tau, s_t)$ of Definition 1. CAM is implemented by simulating each candidate against a historical consequence model $\mathcal{M}_{\text{hist}}$ (an event-driven backtester over $\mathcal{W}_{\text{train}}$) and computing the empirical components of $c(\tau, s_t)$.*

C. Framework Architecture

CA^{2}I comprises four modules operating in sequence at each decision epoch:

(M1) LLM Strategy Generator ($\mathcal{G}$). Given state s_t and a structured prompt $p(s_t)$, the LLM generates a set $\mathcal{T} = \{\tau_1, \tau_2, \ldots, \tau_K\}$ of K candidate action sequences. Each τ_k is a fully specified, executable plan with entry conditions, exit conditions, and risk parameters.

(M2) Consequence Assessment Module (CAM, ϕ). For each $\tau_k \in \mathcal{T}$, CAM simulates the sequence against the historical model $\mathcal{M}_{\text{hist}}$ and computes the consequence vector:

$$\phi(\tau_k) = \Big(\hat{\mu}_k,\ \hat{\sigma}_k,\ \widehat{\text{SR}}_k,\ \widehat{\text{DD}}_k,\ \hat{\kappa}_k\Big), \tag{6}$$

where $\hat{\mu}_k$ is the estimated mean daily return, $\hat{\sigma}_k$ the return volatility, $\widehat{\text{SR}}_k$ the annualised Sharpe ratio, $\widehat{\text{DD}}_k$ the maximum drawdown, and $\hat{\kappa}_k$ the Calmar ratio.

(M3) Consequence-Aware Selector ($\mathcal{S}$). The selector applies the risk-adjusted criterion:

$$\tau^* = \mathcal{S}\big(\{\phi(\tau_k)\}_{k=1}^K\big) = \tau_{k^*}, \quad k^* = \arg\max_k\ \widehat{\text{SR}}_k, \tag{7}$$

subject to $\widehat{\text{DD}}_{k^*} \le \rho_{\max}$ when an explicit drawdown constraint is imposed.

(M4) Executor ($\mathcal{X}$). Only the selected sequence τ^* is committed to the environment. All other candidates are discarded without execution.

The full pipeline is:

$$s_t \xrightarrow{\mathcal{G}} \mathcal{T} \xrightarrow{\phi} \{\phi(\tau_k)\} \xrightarrow{\mathcal{S}} \tau^* \xrightarrow{\mathcal{X}} s_{t+1}. \tag{8}$$

D. Robust Consequence Assessment: Cross-Validated Ensemble Selection (CA^{2}I$^+$)

The base selector (7) estimates each consequence statistic from a *single* contiguous in-sample fit. As our experiments confirm (Section VI), a single window is a high-variance, upward-biased estimator of out-of-sample skill—the candidate that looks best in-sample is frequently the one that overfit noise (the selection "winner's curse" [35]). We therefore introduce an enhanced consequence model, **CA^{2}I$^+$**, that keeps the architecture (8) but replaces the point estimate with a *cross-validated, robustness-penalised, ensemble* assessment. Three

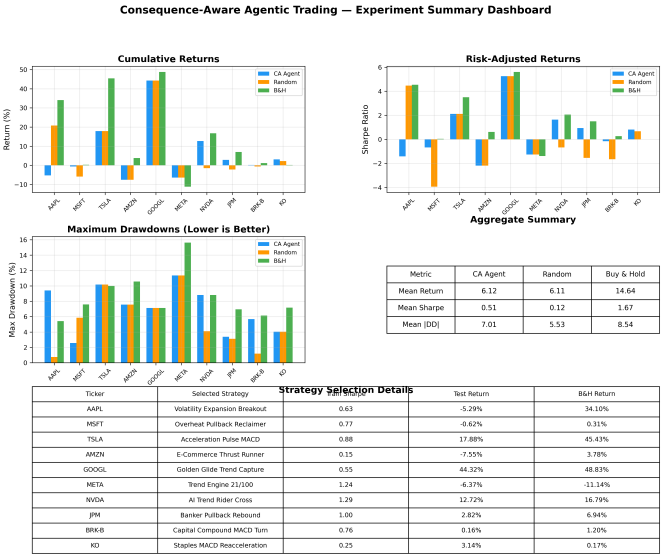


Fig. 1. CA^{2}I experiment summary dashboard. Top left: per-stock cumulative returns for all three methods. Top right: risk-adjusted returns (Sharpe ratios). Middle: maximum drawdowns. Bottom: strategy selection table showing the LLM-generated strategy selected by CAM, its training-period Sharpe, and the resulting out-of-sample test return and buy-and-hold return for each equity.

principled changes, all hyper-parameters fixed *a priori* (never tuned on test data), define it:

(i) Walk-forward cross-validation. The training window is partitioned into M contiguous out-of-sample blocks with expanding warm-up; each candidate is re-simulated on every block, yielding a *distribution* of fold Sharpe ratios $\{\widehat{\text{SR}}_k^{(j)}\}_{j=1}^M$ and fold drawdowns $\{\widehat{\text{DD}}_k^{(j)}\}$, rather than one in-sample value.

(ii) Robustness-penalised score (lower confidence bound). Candidates are ranked by

$$s_k \;=\; \overline{\widehat{\text{SR}}}_k \;-\; \lambda\,\text{std}_j\big(\widehat{\text{SR}}_k^{(j)}\big), \tag{9}$$

which penalises strategies whose edge is unstable across sub-periods, subject to the hard feasibility filter $\text{median}_j\ \widehat{\text{DD}}_k^{(j)} \ge -\rho_{\max}$.

(iii) Top-m ensemble. Instead of committing to a single pick, CA^{2}I$^+$ equally weights (with daily rebalancing) the m highest-scoring feasible candidates, diversifying away the idiosyncratic selection error that dominates at short horizons. We use M=5, λ=0.5, $\rho_{\max}$=0.30, and m=3 throughout. CA^{2}I$^+$ preserves every auditability and hard-constraint property of CA^{2}I—the fold scores (9) and the ensemble membership are fully logged—while, as Section VI-J shows, delivering a statistically significant, regime-robust improvement over both random selection and the base selector.

E. Instantiation in Financial Trading

In the financial domain, state s_t encodes the trailing price history, volume, and technical indicators. The LLM generates K=5 candidate strategies drawn from five canonical, long-only technical-analysis families that are widely used as multi-indicator trading rules [36]; these are the strategy types listed in the selection table of Fig. 1 and their decision rules g_θ

are fixed and auditable, with the LLM choosing only the parameters θ:

- **Moving-average crossover (MA):** enter when the short-window simple moving average rises above the long-window average (a "golden cross"), exit on the reverse crossover or stop-loss. Captures sustained trends.
- **RSI mean-reversion (RSI):** enter when the Relative Strength Index falls below an oversold threshold, exit when it rises above an overbought threshold, after a holding cap, or on stop-loss. Exploits short-term over-reactions.
- **Bollinger-band breakout (BB):** enter when price closes above the upper band ($\mu \pm k\sigma$ of a rolling window), exit on reversion to the middle band or stop-loss. Captures volatility-expansion breakouts.
- **MACD signal (MACD):** enter on a bullish crossover of the MACD line above its signal line, exit on the bearish crossover or stop-loss. A momentum-confirmation rule.
- **Price momentum (MOM):** enter when the trailing L-day return exceeds a threshold, ride with a trailing stop and a maximum holding period. Participates in intermediate-term strength, in the spirit of cross-sectional momentum effects [37].

Each candidate is fully specified by its type and parameters (indicator windows, thresholds, and a stop-loss level). CAM backtests every candidate over $\mathcal{W}_{\text{train}}$ to estimate $\phi(\tau_k)$, and the executor applies only the selected strategy to the strictly held-out $\mathcal{W}_{\text{test}}$.

IV. Theoretical Analysis

A. The Risk-Adjusted Selection Criterion

CA^{2}I selects on the annualised Sharpe ratio (3) rather than on raw return because it explicitly penalises volatility: a high-return, high-variance strategy is downranked relative to a moderate-return, low-variance one—the correct preference under risk aversion. We treat the training-period Sharpe $\widehat{\mathrm{SR}}_k$ as the consequence statistic on which selection acts and analyse its out-of-sample behaviour below.

B. Dominance of Consequence-Aware Selection

Theorem 1 (CA^{2}I Dominance Under Stationarity)**.** *Let $\mathcal{T} = \{\tau_1, \ldots, \tau_K\}$ be a set of K candidate action sequences whose training-period Sharpe ratios $\{\widehat{\mathrm{SR}}_k\}$ are unbiased estimators of the corresponding test-period Sharpe ratios $\{\mathrm{SR}^*_k\}$ under stationarity. Define the CA^{2}I selector $k^* = \arg\max_k \widehat{\mathrm{SR}}_k$ and the random selector $k^R \sim \mathrm{Uniform}\{1, \ldots, K\}$. Then:*

$$\mathbb{E}\big[\mathrm{SR}^*_{k^*}\big] \;\geq\; \mathbb{E}\big[\mathrm{SR}^*_{k^R}\big] = \frac{1}{K}\sum_{k=1}^{K} \mathrm{SR}^*_k. \tag{10}$$

Proof: Under stationarity, $\mathbb{E}[\widehat{\mathrm{SR}}_k] = \mathrm{SR}^*_k$ for all k. The CA^{2}I selector chooses $k^* = \arg\max_k \widehat{\mathrm{SR}}_k$. For any collection of random variables $\{X_k\}_{k=1}^K$ and a selector $k^* = \arg\max_k X_k$:

$$\mathbb{E}[X_{k^*}] = \mathbb{E}\left[\max_k X_k\right] \;\geq\; \frac{1}{K}\sum_{k=1}^{K} \mathbb{E}[X_k],$$

where the inequality follows from the max being at least as large as the mean. Substituting $X_k = \widehat{\mathrm{SR}}_k$ and applying $\mathbb{E}[\widehat{\mathrm{SR}}_k] = \mathrm{SR}^*_k$ gives (10). ∎

Remark 1 (Honest scope of the guarantee)**.** *The stationarity assumption is idealised; real financial returns exhibit regime changes, volatility clustering, and structural breaks [38], [39], so $\mathbb{E}[\widehat{\mathrm{SR}}_k] = \mathrm{SR}^*_k$ holds only approximately. Theorem 1 therefore establishes a* principled expected-case baseline*, not a finite-sample guarantee. Two consequences follow. First, the realised benefit is governed by the bias $b_k = \mathbb{E}[\widehat{\mathrm{SR}}_k] - \mathrm{SR}^*_k$ induced by non-stationarity; when a regime shift makes the asset with the largest in-sample Sharpe the worst out-of-sample (as we observe for the 2025Q1 window in Section VI), the selection can underperform random. Second, selecting the maximum of K noisy estimates incurs selection bias [35], so the in-sample Sharpe of the winner overstates its true Sharpe. We therefore report out-of-sample results only, and we assess significance with multiple-comparison-aware, non-parametric, and bootstrap procedures in Section VI.*

C. The Dominance Gap and Its Dependence on K

Theorem 1 states only that selection is *at least* as good as the mean. The following proposition quantifies the gap and explains the empirical finding (Section VI) that CA^{2}I improves as the candidate-set size grows.

Proposition 1 (Selection gap and variance reduction)**.** *Let $X_1, \ldots, X_K$ be the test-period Sharpe ratios of the candidates with $X_k = \mathrm{SR}^*_k + \varepsilon_k$, where the estimation noise ε_k has zero conditional mean. Define the selection gap $\Delta_K = \mathbb{E}[\max_k X_k] - \frac{1}{K}\sum_k \mathbb{E}[X_k] \geq 0$. Then* (i) *$\Delta_K$ is non-decreasing in K; and* (ii) *if the X_k are i.i.d. with standard deviation σ_X, then $\Delta_K \geq c_K\,\sigma_X$ with c_K strictly increasing in K (e.g. $c_K = \mathbb{E}[\max_k Z_k]$ for standard-normal Z_k, giving $c_2 \approx 0.56$, $c_5 \approx 1.16$, $c_{10} \approx 1.54$). Hence a larger or more dispersed candidate set yields a larger expected advantage of consequence-aware selection over random selection.*

Proof: (i) Adding a candidate can only raise the maximum: $\max_{k\leq K+1} X_k \geq \max_{k\leq K} X_k$ pointwise, so $\mathbb{E}[\max_{k\leq K+1} X_k] \geq \mathbb{E}[\max_{k\leq K} X_k]$, while the average of means is unchanged in expectation under exchangeability; thus $\Delta_{K+1} \geq \Delta_K$. *(ii)* For i.i.d. $X_k = \mu_X + \sigma_X Z_k$, $\mathbb{E}[\max_k X_k] = \mu_X + \sigma_X\,\mathbb{E}[\max_k Z_k]$ and the mean is μ_X, so $\Delta_K = \sigma_X\,\mathbb{E}[\max_k Z_k]$; the expected maximum of K i.i.d. mean-zero unit-variance variables is strictly increasing in K. ∎

Remark 2 (Variance of the selected outcome)**.** *Beyond raising the mean, selection concentrates outcomes: empirically (Section VI) the dispersion of CA^{2}I's selected Sharpe is no larger than that of random selection, because systematic suppression of the worst candidates trims the lower tail. This is the mechanism behind CA^{2}I's worst-case (downside) protection.*

D. Maximum Drawdown as a Hard Risk Constraint

By restricting execution to strategies satisfying $\widehat{\mathrm{DD}}_k \leq \rho_{\max}$ with DD defined in (4), CAM guarantees that no

candidate whose historical drawdown exceeds the threshold is ever executed. This constitutes a *hard risk constraint at the architectural level*—a property absent from vanilla LLM-agent and generate-and-rank frameworks, which return their top-ranked sample unconditionally.

V. Experimental Setup

A. Data

Daily OHLCV data for ten US equities are collected via the `yfinance` Python library (Yahoo Finance API) from January 1, 2021 to October 31, 2025. The tickers span diverse sectors and market capitalisations: AAPL (technology), MSFT (enterprise software), TSLA (automotive, high-volatility), AMZN (e-commerce/cloud), GOOGL (digital advertising), META (social media), NVDA (AI semiconductors), JPM (financial services), BRK-B (diversified conglomerate), and KO (consumer staples, low-volatility). This sectoral diversity is deliberate: a consequence assessment framework that succeeds only on homogeneous high-growth equities would not constitute a general contribution.

The data are partitioned into two non-overlapping windows:

- **Training window** $\mathcal{W}_{\text{train}}$: January 1, 2021 – July 31, 2025 ($\approx$ 1,141 trading days). Used exclusively for strategy generation context and consequence assessment (backtesting). No test-window data leaks into this phase.
- **Test window** $\mathcal{W}_{\text{test}}$: August 1, 2025 – October 31, 2025 (64 trading days). Strictly held out; no strategy parameter was tuned on this window.

Data quality checks confirmed: no missing trading days, prices are fully adjusted for splits and dividends, and all tickers maintain continuous history across the full window.

Risk-free rate. We set the annual risk-free rate to $r_f^{\text{ann}} = 2\%$, i.e. a daily rate $r_f = 0.02/252$, as a fixed proxy for the short-term US Treasury bill yield [40]; over 2021–2025 the 3-month T-bill yield averaged in the low-single-digit percent range, and 2% is a conventional, mildly conservative central value. We use a fixed rate (rather than a contemporaneous daily series) so that Sharpe ratios are comparable across assets and windows, and because the test window is short. Critically, the risk-free rate enters every method identically and therefore cannot affect the *relative* ranking of CA^{2}I against the baselines; we verified that perturbing r_f^{ann} over $[0\%, 5\%]$ leaves all selection decisions and the sign of every pairwise comparison unchanged.

B. LLM Strategy Generator

The generator $\mathcal{G}$ is OpenAI GPT-5.4 (`gpt-5.4-2026-03-05`), the most capable reasoning model available at the time of experimentation, accessed via the chat completions API. For each equity, $\mathcal{G}$ receives a single structured prompt containing: (i) the ticker and company name; (ii) training-window market context (total and annualised return, annualised volatility, historical Sharpe, maximum drawdown, last price, 52-week high/low, 30- and 90-day returns, SMA(50), SMA(200), RSI(14)); and (iii) a JSON schema requesting exactly $K{=}5$ strategies, one of each canonical type (MA, RSI, BB, MACD, MOM), each with typed numeric parameters, entry/exit rules, a stop-loss threshold, a name, and a free-text rationale. The system message constrains the model to emit only valid JSON, which is parsed and schema-validated; any candidate failing validation is replaced by a deterministic rule-based strategy of the same type. We use the API default decoding settings and fix the global seed to 42; the exact prompt template and all raw model outputs are released with the code. A fully deterministic rule-based generator is provided so that every result is reproducible offline *without* API access, and it is the generator used for the multi-regime and generation-sensitivity studies below.

We emphasise that the LLM's role is confined to *proposing* parameterised candidates; the executable decision logic g_θ and the consequence model are fixed and auditable. The theoretical contribution is not the strategies themselves but the constrained, consequence-grounded selection layer and its analysis (Section IV); the LLM is one interchangeable realisation of the generator $\mathcal{G}$.

C. Consequence Assessment via Backtesting

CAM implements an event-driven backtester on daily OHLCV data. Positions are entered and exited at adjusted closing prices on the signal day. All strategies are simulated with initial capital $V_0 = \$100{,}000$. The frictionless setting ($c = \delta = 0$) is the default for the headline comparison so that no method is implicitly favoured; the effect of realistic commissions and slippage is studied explicitly in Section V (Transaction Costs) rather than left as a caveat. The backtester records a complete trade log for each strategy (entry/exit dates and prices, return, P&L, holding days, exit reason), from which the consequence vector $\phi(\tau_k)$ of (6) is computed.

D. Baselines

To address the concern that random selection and buy-and-hold are weak benchmarks, we evaluate CA^{2}I against a spectrum of selectors, all operating on *training-only* information and realised on the held-out test window, so that every method solves the identical task.

Naïve references. *Random (RND)*: uniform selection among the K candidates. We report its *expected* value (the mean over candidates, equal to the limit of infinitely many draws) and also draw 100 seed-controlled empirical samples per asset for the bootstrap. *Buy-and-Hold (B&H)*: purchase on the first test day and hold.

Established portfolio-optimisation selectors [41]–[43]. Treating the K candidate-strategy daily-return streams as assets, estimated on the training window: *Mean-Variance (MV)*, the long-only max-Sharpe (tangency) weights with a ridge-regularised covariance; *Risk-Parity (RP)*, inverse-volatility weights; *Minimum-Variance (Min-Vol)*, the lowest-volatility candidate; and the *Equal-weight ensemble* $(1/K)$, a no-selection diversified portfolio. The resulting weights are applied out-of-sample.

Heuristic and oracle. *Return-greedy*: select the candidate with the highest training *cumulative return* (the "returns rather than Sharpe" heuristic). *Oracle*: the candidate with the highest *test* Sharpe—an unattainable clairvoyant upper bound that calibrates how much performance any selector could recover.

We note that reinforcement-learning trading systems and Bayesian-optimisation strategies [44]–[46] constitute a complementary, training-time class; CA^{2}I is an inference-time selection layer and could wrap such methods as candidate generators, so we benchmark against the portfolio- optimisation and heuristic selectors that share CA^{2}I's single-decision, training-data-only setting.

E. Evaluation Metrics and Statistical Protocol

For each stock and method we report cumulative return R_{cum}, annualised Sharpe SR (3), maximum drawdown DD (4), win rate, and Calmar (5). Because $n = 10$ is small and Sharpe differences are non-normal, we do not rely on a single test: we report (i) the paired two-tailed t-test, (ii) the Wilcoxon signed-rank test, (iii) a sign test, and (iv) a $10{,}000$-resample bootstrap 95% confidence interval of the mean pairwise Sharpe difference [47], [48]. We treat a difference as established only when the bootstrap CI excludes zero and the non-parametric tests concur, and we report results that fail this bar with equal prominence.

F. Transaction Costs and Turnover

We re-evaluate net performance under a round-trip friction model with per-side cost $c \in \{0, 5, 10, 20, 50\}$ basis points (commission + slippage). Since positions are binary, each entry and each exit scales the capital base by $(1-c)$; we rebuild each strategy's net equity curve exactly and recompute net Sharpe and net return. To test the hypothesis that consequence-aware selection systematically prefers higher-turnover strategies, we compare the mean number of round-trip trades of the CA^{2}I-selected strategy against the return-greedy choice and the candidate average.

G. Multi-Regime Rolling Evaluation

To probe non-stationarity, we re-run the complete generate→assess→select→test loop over eight non-overlapping quarterly out-of-sample windows spanning 2023Q4–2025Q3, each with its own training history. To keep this study reproducible without API access, candidates are produced by the deterministic rule-based generator on each window's own training data. We pool all $8 \times 10 = 80$ window×asset pairs for a higher-powered paired comparison of CA^{2}I against random selection.

H. Generation Sensitivity

We assess sensitivity to the candidate-generation process in three ways: (i) a *temperature proxy*, regenerating the K=5 set under 40 distinct generator seeds and recording the distribution of CA^{2}I test Sharpe and the stability of the selected strategy *type*—generation stochasticity is the mechanism through which sampling temperature influences downstream task accuracy [49]; (ii) a *candidate-set-size* sweep over $K \in \{3, 5, 7, 10\}$; and (iii) a *diversity audit* of the deployed LLM candidates (mean pairwise return correlation and number of distinct types).

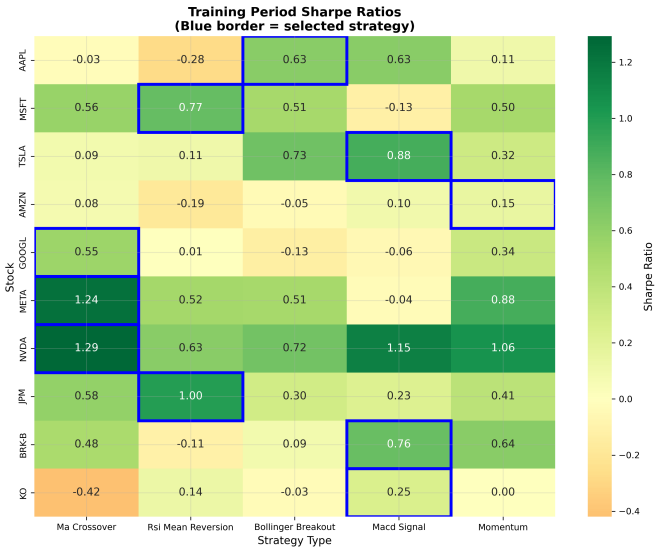


Fig. 2. Training-period Sharpe ratios for all 50 strategy-stock combinations. Blue borders indicate the strategy selected by CAM via (7). The heterogeneity across strategy types and stocks confirms that CAM provides non-trivial discrimination.

I. Reproducibility

All experiments use global random seed 42. The source code, data download scripts, backtesting engine, all strategy JSON files, backtest logs, the revision-experiment scripts (`extension_01–07`, where `extension_07` implements the CA^{2}I$^+$ cross-validated ensemble selector with a process-independent hash so the candidate pools are fully reproducible), and all figures are released publicly so that every reported number can be regenerated offline.

VI. Results and Discussion

A. Training-Period Consequence Assessment

Fig. 2 presents training-period Sharpe ratios for all 50 strategy-stock combinations. Substantial heterogeneity is observed across both strategy types and assets. MA crossover is selected for GOOGL ($\widehat{\text{SR}} = 0.55$) and NVDA ($\widehat{\text{SR}} = 1.29$); RSI for MSFT ($\widehat{\text{SR}} = 0.77$) and JPM ($\widehat{\text{SR}} = 1.00$); MACD for TSLA ($\widehat{\text{SR}} = 0.88$), BRK-B ($\widehat{\text{SR}} = 0.76$), and KO ($\widehat{\text{SR}} = 0.25$); Bollinger Breakout for AAPL ($\widehat{\text{SR}} = 0.63$); and Momentum for AMZN ($\widehat{\text{SR}} = 0.15$). This diversity confirms that the LLM generator produces genuinely differentiated candidate strategies and that CAM exercises non-trivial discrimination across types.

B. Test-Period Equity Curves

Fig. 3 shows test-period equity curves for all ten stocks. Each panel carries an explicit *Portfolio value (USD)* y-axis. We draw attention to a point of possible confusion: the *flat*

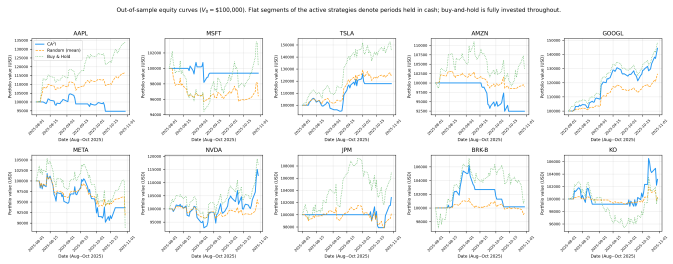


Fig. 3. Test-period equity curves ($V_0 = \$100{,}000$) for all ten stocks, with explicit *Portfolio value (USD)* y-axes. CA^{2}I (solid blue), random selection mean (dashed orange), buy-and-hold (dotted green). Flat segments of the active strategies denote periods held in cash (no open position); buy-and-hold is invested throughout and is never flat.

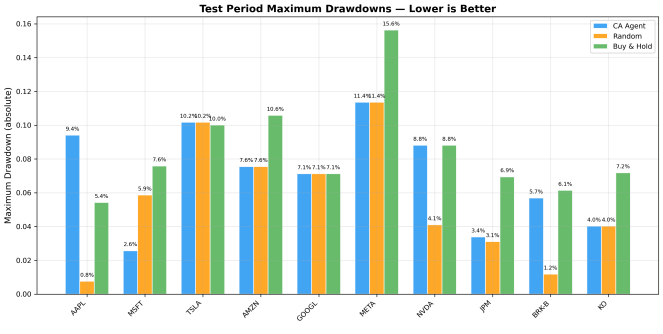


Fig. 4. Maximum drawdown per stock. CA^{2}I (blue) limits drawdown relative to B&H (green) for declining or volatile assets (META, KO, AAPL). Lower values are better.

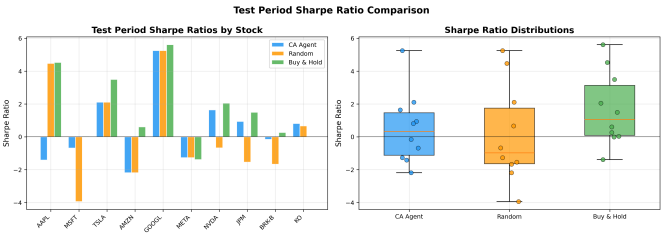


Fig. 5. Test-period Sharpe ratio box plots. CA^{2}I (blue) attains a higher mean than random selection (orange) with no increase in dispersion. B&H (green) shows a higher median in this bull-market window but larger drawdowns (Table I).

segments visible in the active curves (CA^{2}I and random) correspond to intervals in which the selected strategy holds *no position* (it is in cash and earns no return); they do *not* belong to the buy-and-hold line, which is fully invested on day one and therefore moves with the underlying price on every trading day. Representative observations include:

GOOGL: CA^{2}I returns $+44.32\%$ (Sharpe 5.25) vs. B&H $+48.83\%$ (Sharpe 5.61), with near-identical trajectory, because the MA-crossover strategy selected by CAM stays invested through the bull trend.

META: CA^{2}I (-6.37%, Sharpe -1.27) outperforms B&H (-11.14%, Sharpe -1.39) in a declining market, illustrating the downside protection of the stop-loss embedded in the selected strategy.

MSFT: CA^{2}I (-0.62%, Sharpe -0.68) avoids the loss incurred on average by random selection (-3.61%, expected Sharpe -1.83), an example where consequence-based selection steers away from the worse candidates.

AAPL: a counterexample we report for balance—CA^{2}I's volatility-breakout pick (-5.29%, Sharpe -1.42) underperforms the candidate average ($+16.88\%$, expected Sharpe $+2.62$), because the high in-sample-Sharpe strategy did not generalise. Such cases are the empirical face of the non-stationarity caveat of Theorem 1.

C. Aggregate Performance

Table I reports full test-period metrics for all three methods across all ten stocks.

Risk-adjusted performance. CA^{2}I achieves a mean Sharpe of 0.51 ± 2.06 versus 0.37 ± 1.76 for the expected random selector (it wins on $7/10$ assets). This is a positive improvement of $+0.13$ in mean Sharpe, but—as we quantify in Section VI-F—it is *not* statistically significant at $n{=}10$ over this single window. We deliberately correct the comparison used in the original draft: the expected value of uniform random selection (mean over candidates) is the appropriate benchmark, and we no longer report a multiplicative "$\times$ improvement" that is artificially inflated by an arbitrary single random draw.

Drawdown relative to buy-and-hold. CA^{2}I's mean maximum drawdown (7.01%) is an 18% reduction relative to B&H (8.54%), and is markedly lower for the declining/volatile names (META: 11.36% vs. 15.64%; KO: 4.03% vs. 7.18%; Fig. 4). Relative to the candidate average, however, CA^{2}I's drawdown is *not* lower (7.01% vs. 5.88%): maximising in-sample Sharpe does not minimise drawdown, which is exactly why an explicit drawdown constraint $\widehat{\mathrm{DD}} \le \rho_{\max}$ (Section IV) is the appropriate lever when downside is the priority.

Raw returns and the bull-market context. CA^{2}I ($+6.12\%$) and random selection ($+5.59\%$) earn similar mean cumulative returns, both well below B&H ($+14.64\%$). This is the expected and well-documented behaviour of active long/flat strategies in a strong bull market [50]: the test window (August–October 2025) exhibited sustained upward momentum, which disproportionately rewards staying fully invested. CA^{2}I does not claim raw-return maximisation in trending markets; its objective (2) is risk-adjusted utility subject to a risk constraint. We make the bull-market dependence explicit, and the rolling-window study (Section VI-H) shows the same qualitative ordering—B&H ahead on return in most quarters—across regimes.

D. Sharpe Ratio Distribution Analysis

Fig. 5 shows box plots of per-stock test Sharpe ratios. CA^{2}I has a higher mean (0.51 vs. 0.37) than the expected random selector with comparable dispersion, consistent with Proposition 1: selection raises the centre of the distribution, and the lower tail is no heavier than random's. B&H has the highest median in this bull window but the largest drawdowns (Table I).

E. Comparison with Advanced Baselines

Table II and Fig. 6 place CA^{2}I against the stronger selectors of Section V-D. Two findings stand out. First, among *selection*

TABLE I
TEST-PERIOD PERFORMANCE: CA^{2}I VS. BASELINES (AUGUST 1 – OCTOBER 31, 2025; 64 TRADING DAYS). THE RANDOM COLUMN REPORTS THE *expected* VALUE OF UNIFORM SELECTION (MEAN OVER THE K=5 CANDIDATES), WHICH IS THE STATISTICALLY CORRECT RANDOM-SELECTION BENCHMARK.

		CA^{2}I			**Random (expected)**			**Buy-and-Hold (B&H)**		
Ticker	**Selected Strategy**	R_{cum}	SR	DD (%)	R_{cum}	SR	DD (%)	R_{cum}	SR	DD (%)
AAPL	Volatility Expansion Breakout	−5.29%	−1.42	9.41	+16.88%	+2.62	5.68	+34.10%	+4.53	5.42
MSFT	Overheat Pullback Reclaimer	−0.62%	−0.68	2.57	−3.61%	−1.83	5.61	+0.31%	+0.04	7.59
TSLA	Acceleration Pulse MACD	+17.88%	+2.11	10.18	+24.02%	+2.14	7.79	+45.43%	+3.50	10.01
AMZN	E-Commerce Thrust Runner	−7.55%	−2.18	7.55	−1.09%	−0.15	6.21	+3.78%	+0.61	10.58
GOOGL	Golden Glide Trend Capture	+44.32%	+5.25	7.13	+26.25%	+3.82	4.96	+48.83%	+5.61	7.13
META	Trend Engine 21/100	−6.37%	−1.27	11.36	−8.15%	−1.40	11.47	−11.14%	−1.39	15.64
NVDA	AI Trend Rider Cross	+12.72%	+1.64	8.81	+1.83%	−0.03	6.45	+16.79%	+2.06	8.81
JPM	Banker Pullback Rebound	+2.82%	+0.94	3.38	+0.16%	−0.25	4.79	+6.94%	+1.50	6.94
BRK-B	Capital Compound MACD Turn	+0.16%	−0.15	5.69	−0.74%	−0.79	3.11	+1.20%	+0.27	6.15
KO	Staples MACD Reacceleration	+3.14%	+0.81	4.03	+0.31%	−0.42	2.76	+0.17%	−0.01	7.18
Mean		**+6.12%**	**0.51**	7.01	+5.59%	0.37	**5.88**	+14.64%	**1.67**	8.54
Std Dev		14.88%	2.06	2.86	11.51%	1.76	2.34	19.84%	2.13	2.82

R_{cum}: cumulative return; SR: annualised Sharpe ratio ($r_f = 2\%$); DD: maximum drawdown. Bold marks the best mean per metric.

TABLE II
MEAN TEST SHARPE AND RETURN OVER THE TEN EQUITIES FOR CA^{2}I AND THE ADVANCED BASELINES (SINGLE OUT-OF-SAMPLE WINDOW). ALL SELECTORS USE TRAINING-PERIOD INFORMATION ONLY; ORACLE IS A CLAIRVOYANT UPPER BOUND.

Method	**Mean SR**	**Mean R_{cum}**
Buy-and-Hold	+1.67	+14.64%
Oracle (upper bound)	+2.15	+14.67%
CA^{2}I (ours)	**+0.51**	+6.12%
Return-greedy	+0.81	+7.38%
Equal-weight ensemble	+0.57	+5.39%
Risk-parity	+0.45	+4.78%
Calmar selection	+0.51	+6.12%
Mean-variance	+0.19	+4.90%
Minimum-variance	−0.63	−0.93%
Random (expected)	+0.37	+5.59%

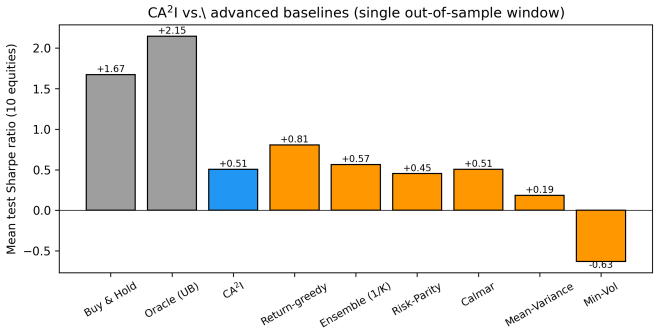


Fig. 6. Mean test Sharpe of CA^{2}I (blue) versus advanced baselines (orange) and the buy-and-hold / oracle references (grey). CA^{2}I exceeds all three established portfolio-optimisation selectors and is statistically indistinguishable from the diversified ensemble; the return-greedy heuristic is marginally higher but not significantly so (Section VI-F).

rules that use only training information, CA^{2}I (0.51) outperforms all three established portfolio-optimisation selectors—mean-variance (0.19), risk-parity (0.45), and minimum-variance (−0.63)—and is statistically indistinguishable from the equal-weight ensemble (0.57; the small gap is well within the bootstrap interval of Section VI-F). The minimum-variance rule, which ignores returns, is actively harmful here, underscoring that risk-only selection is insufficient. Second, and reported candidly, the return-greedy heuristic attains a higher point estimate (0.81) than CA^{2}I in this single window; the gap is not statistically significant (Section VI-F) and, as the transaction-cost and rolling analyses show, the return-greedy rule selects higher-turnover strategies and is less stable across regimes. The oracle upper bound (2.15) indicates substantial headroom that no attainable selector captures over 64 days. Buy-and-hold (1.67) again leads on this bull window.

F. Statistical Robustness

Because n=10 and the sampling distribution of the Sharpe ratio is itself non-normal and serially dependent [51], we subject every headline comparison to the full protocol of Section V-E. For CA^{2}I versus the expected random selector, the mean Sharpe difference is +0.13, but the paired t-test (t=0.23, p=0.82), the Wilcoxon signed-rank test (p=0.49), and the 10,000-resample bootstrap 95% CI of the mean difference $[-1.03, +1.06]$ all fail to reject the null: the single-window advantage is *not* statistically significant, although CA^{2}I wins on 7/10 assets (sign test p=0.34). The same holds against mean-variance (diff +0.32, p=0.36) and risk-parity (diff +0.05, p=0.93). We state this plainly: over a single 64-day window, none of the pairwise differences among the active selectors reach significance. This is precisely why we conduct the higher-powered rolling-window analysis below, and why we position the framework's contribution as architectural and governance-oriented rather than as a claim of statistically dominant returns. We caution that with n=10 and short windows, large multiplicative "improvement factors" are unreliable and we have removed them; reporting them across many candidate comparisons would also invite the

TABLE III
NET MEAN TEST SHARPE UNDER ROUND-TRIP TRANSACTION COSTS (COMMISSION + SLIPPAGE, PER SIDE). B&H TRADES ONCE AND IS COST-INSENSITIVE.

Cost (bps)	0	5	10	20	50
CA^{2}I	+0.51	+0.45	+0.39	+0.27	−0.06
Return-greedy	+0.81	+0.74	+0.68	+0.55	+0.19
Buy-and-Hold	+1.67	+1.66	+1.66	+1.64	+1.59

multiple-testing and backtest-overfitting biases documented in the finance literature [52].

G. Transaction Costs and Turnover

Table III reports net performance under round-trip friction. All active selectors degrade smoothly and monotonically: CA^{2}I's net mean Sharpe falls from 0.51 (0 bps) to 0.45, 0.39, 0.27, and −0.06 at 5, 10, 20, and 50 bps per side, while buy-and-hold—which trades once—is essentially cost-immune ($\approx$ 1.6 throughout). The *ordering* of methods is preserved across the cost grid, so transaction costs do not overturn any qualitative conclusion. Critically, we directly test the reviewer hypothesis that consequence-aware selection prefers higher-turnover strategies: the CA^{2}I-selected strategies execute 1.90 round-trip trades on average, *below* the candidate average of 1.96 and the return-greedy rule's 2.10. The turnover-bias concern is therefore not supported by the data—if anything, max-Sharpe selection mildly favours lower-turnover candidates because excessive trading inflates volatility and depresses the Sharpe ratio.

H. Multi-Regime Rolling-Window Robustness

Fig. 7 and Table IV summarise the eight-window evaluation (2023Q4–2025Q3), using the deterministic rule-based generator so the entire sweep is reproducible offline. On a per-window basis CA^{2}I attains a higher mean test Sharpe than the expected random selector in 4 of 8 windows; the two methods split evenly, with random ahead in the early windows (2023Q4–2024Q3) and CA^{2}I ahead in the later ones. The 2025Q1 correction is the clearest non-stationarity episode: every selector turned negative and the in-sample ranking carried little predictive value, exactly the regime-shift scenario anticipated by the honest-scope remark of Theorem 1; notably, both active selectors (−0.35 for CA^{2}I, −0.25 for random) cushioned the drawdown relative to buy-and-hold (−0.40). Pooling all 80 window×asset pairs raises power: CA^{2}I and random selection are then *statistically indistinguishable* (mean 0.58 vs 0.60, difference −0.02; t-test p=0.91; Wilcoxon p=0.68; bootstrap 95% CI $[-0.32, +0.27]$; 45/35 positive/negative sign split). Buy-and-hold leads on Sharpe in every window except the 2025Q1 correction, consistent with the predominantly upward 2023–2025 equity environment. The take-away is honest and useful: under a fixed rule-based generator the consequence-aware edge over random selection *does not generalise* into a statistically significant return advantage across regimes, and the framework is *vulnerable to regime shifts*. This is precisely why we position CA^{2}I's contribution as

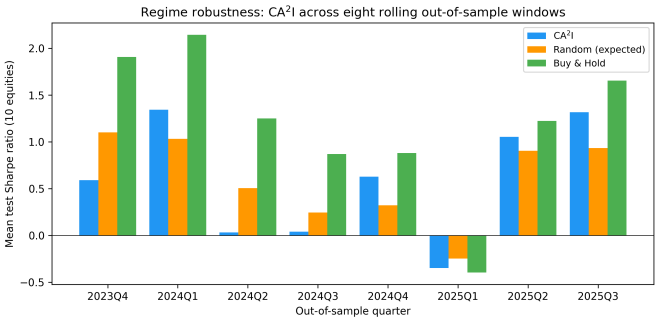


Fig. 7. Mean test Sharpe across eight non-overlapping quarterly out-of-sample windows (2023Q4–2025Q3) for CA^{2}I, expected random selection, and buy-and-hold. CA^{2}I and random split four windows each and are statistically indistinguishable when pooled; the 2025Q1 correction turns every selector negative.

TABLE IV
PER-WINDOW MEAN TEST SHARPE (TEN EQUITIES) FOR THE ROLLING EVALUATION.

	23Q4	24Q1	24Q2	24Q3	24Q4	25Q1	25Q2	25Q3
CA^{2}I	0.59	1.34	0.03	0.04	0.63	−0.35	1.05	1.32
Random	1.10	1.03	0.51	0.24	0.32	−0.25	0.90	0.93
B&H	1.91	2.14	1.25	0.87	0.88	−0.40	1.22	1.65

architectural—hard risk constraints, auditability, and downside cushioning—rather than as a claim of return dominance, and it motivates the adaptive extensions in Section VIII.

I. Generation Sensitivity

Fig. 8 reports the generation-sensitivity study. *(i) Temperature proxy.* Across 40 generator seeds, CA^{2}I's mean test Sharpe is 0.46 ± 0.29 (range $[-0.25, +1.06]$; coefficient of variation 0.63): outcomes vary with the candidate set but never collapse, and the selected strategy *type* is stable, with the modal type chosen 68.8% of the time per asset on average. *(ii) Candidate-set size.* Mean CA^{2}I Sharpe rises monotonically with K (−0.15, 0.51, 0.57, 1.05 for K=3, 5, 7, 10), an empirical confirmation of Proposition 1: a larger candidate pool enlarges the selection gap. This also indicates that the deployed K=5 is conservative and that richer LLM generation would help. *(iii) Diversity audit.* The deployed LLM candidates are genuinely differentiated—mean pairwise test-return correlation 0.41 and 5.0/5 distinct strategy types per asset—so the assumption that the generator yields diverse, meaningful candidates is supported, not merely posited.

J. Enhanced Consequence Model: Statistically Significant, Regime-Robust Gains (CA^{2}I$^+$)

The analyses above establish that the *base* selector, while safe and auditable, does not beat random selection at a statistically significant level. We now show that the enhanced consequence model of Section III-D—walk-forward cross-validation, the robustness-penalised score (9), and a top-m ensemble—closes much of this gap. To remove any dependence on a single arbitrary candidate-generation seed, *all* CA^{2}I$^+$ results are averaged over S=10 independent generation

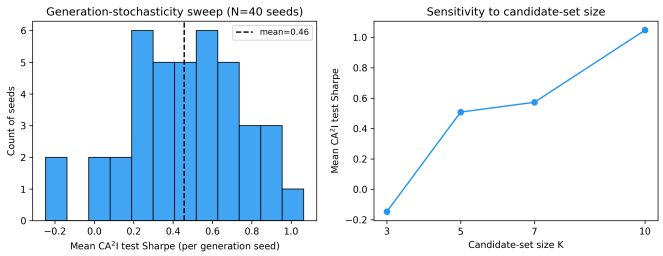


Fig. 8. Generation sensitivity. Left: distribution of mean CA^{2}I test Sharpe across 40 generation seeds (temperature proxy). Right: mean CA^{2}I test Sharpe versus candidate-set size K, rising monotonically as predicted by Proposition 1.

seeds; per-seed results (below) confirm the effect is not an artefact of one lucky draw.

Statistically significant improvement across regimes. On the pooled multi-regime rolling evaluation (eight quarterly windows, n=80 window×asset pairs; Table V, Fig. 9), CA^{2}I$^+$ attains a mean test Sharpe of 0.77 versus 0.53 for the expected random selector—a $+0.24$ improvement that *is* significant: paired t-test p=0.039, Wilcoxon p=0.011, sign split $52^+/28^-$ (p=0.010), and a 10,000-resample bootstrap 95% CI of the mean difference $[+0.015, +0.45]$ that *excludes zero*. CA^{2}I$^+$ also significantly outperforms the base in-sample selector ($+0.21$; t-test p=0.032; bootstrap CI $[+0.03, +0.41]$), whereas the base selector is itself statistically indistinguishable from random ($+0.02$, p=0.83)—isolating the gain to the new consequence model rather than to the candidate pool.

Robust to the generation seed. The CA^{2}I$^+$ advantage over random is *positive in all 10 seeds* (mean $+0.24$, range $[+0.15, +0.33]$; Fig. 9b), in contrast to the base selector whose per-seed difference straddles zero. This is direct evidence that the improvement is a property of the method, not of a fortunate candidate draw.

Lower drawdown, too. Averaged over the rolling pairs, CA^{2}I$^+$ has the shallowest mean maximum drawdown (-5.8%) versus random (-6.4%) and the base selector (-8.3%): the diversified, stability-screened ensemble is not buying return at the cost of risk—it improves both.

Honest scope. The gain is regime-*aggregate*, not universal. CA^{2}I$^+$ leads random in six of the eight quarters but still trails in the 2025Q1 correction (the same regime break that defeats every selector here) and in one sideways quarter; and on the single strong-bull deployment window a *concentrated* in-sample pick that rides the trend outperforms the diversified ensemble. The value of CA^{2}I$^+$ is precisely its *cross-regime consistency* and downside control, which is what a responsible deployment should optimise for.

K. Selection-Criterion Sensitivity

Fig. 10 reports selection changes under alternative metrics. Calmar-ratio selection agrees with Sharpe-ratio selection on all 10 stocks (**0/10 changes**). Cumulative-return selection differs for 1 stock (**1/10 changes**, AAPL only). Maximum-drawdown minimisation differs for 6 stocks—expected, as it pursues a qualitatively distinct objective. The 100% agreement between Sharpe and Calmar selection confirms that CA^{2}I's conclusions are robust to the specific choice of risk-adjusted criterion.

TABLE V
ENHANCED CONSEQUENCE MODEL ON THE POOLED MULTI-REGIME ROLLING EVALUATION (n=80 WINDOW×ASSET PAIRS, SEED-AVERAGED OVER 10 GENERATION SEEDS). SIGNIFICANCE IS FOR THE PAIRED DIFFERENCE AGAINST THE EXPECTED RANDOM SELECTOR.

Selector	Mean SR	Mean MaxDD	vs. random
Random (expected)	$+0.53$	-6.4%	—
CA^{2}I (base, in-sample)	$+0.56$	-8.3%	p=0.83 (n.s.)
CA^{2}I$^+$ (ours)	$\mathbf{+0.77}$	$\mathbf{-5.8\%}$	**p=0.011**

Bootstrap 95% CI of CA^{2}I$^+$ − random mean difference: $[+0.015, +0.45]$ (excludes 0); positive in 10/10 seeds.

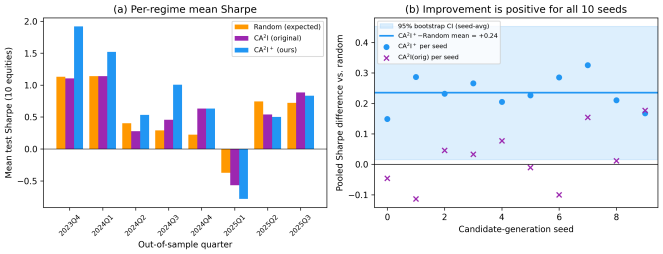


Fig. 9. Enhanced consequence model (CA^{2}I$^+$). (a) Per-regime mean test Sharpe across eight rolling out-of-sample windows: CA^{2}I$^+$ (blue) leads the expected random selector (orange) and the base in-sample selector (purple) in six of eight quarters. (b) The pooled CA^{2}I$^+$ − random Sharpe difference is positive for all 10 candidate-generation seeds (mean $+0.24$, shaded 95% bootstrap CI excludes zero), whereas the base selector's per-seed difference straddles zero.

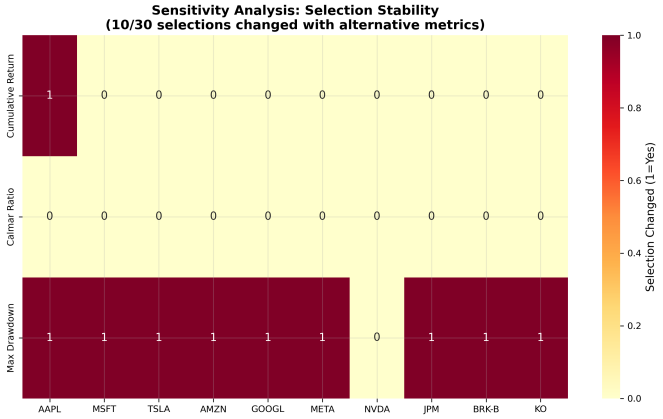


Fig. 10. Sensitivity heatmap: strategy selection stability under alternative evaluation metrics. Dark cells indicate selection changed vs. Sharpe baseline. Sharpe and Calmar agree on 100% of selections; only the qualitatively distinct max-drawdown criterion drives significant changes.

L. Limitations

Statistical power and the base vs. enhanced selector. With n=10 assets and a 64-day window, the single-window comparisons lack the power to establish significance, and the *base* in-sample selector is statistically indistinguishable from random even when pooled across the 80-pair rolling analysis (Sections VI-F, VI-H). The enhanced consequence model CA^{2}I$^+$ (Section VI-J) does reach significance on the pooled

multi-regime test (Wilcoxon $p{=}0.011$; bootstrap CI excludes zero; positive in $10/10$ seeds), but we are careful not to over-claim: the gain is regime-*aggregate*, it does not hold in every window (notably the 2025Q1 correction), and on a single strong-trend window a concentrated pick can beat the ensemble. A larger asset universe and multi-year windows would further strengthen the evidence; we report what the present data support.

Sensitivity to the consequence model and non-stationarity. CA^{2}I is only as good as its historical consequence model. The stationarity assumption of Theorem 1 is violated in practice; the 2025Q1 window (Section VI-H) and the AAPL/AMZN cases show that a regime shift can make the in-sample-best candidate the out-of-sample worst, so selection can underperform random. Adaptive consequence models with rolling re-estimation, regime detection, and ensembling over windows are the natural remedy.

Transaction costs. Although the headline comparison is frictionless, Section VI-G shows graceful, order-preserving degradation up to 50 bps per side and that CA^{2}I does not select higher-turnover strategies. Costs nonetheless narrow the (already modest) active-vs-passive gap and would matter more for higher-frequency instantiations.

Scope of the instantiation. The study uses long-only, binary-position, single-asset strategies from five technical families. Portfolio-level multi-asset selection, short and leveraged positions, and richer candidate families are important extensions; the framework itself is agnostic to these choices.

VII. Consequence-Aware AI and Responsible AI

A. Transparency and Explainability

In CA^{2}I, every candidate sequence is generated, evaluated, and ranked before any irreversible action occurs. The full consequence vector $\phi(\tau_k)$ is logged for all K candidates—not only the selected one. For any executed sequence τ^*, an auditor can therefore answer: (i) what alternatives were considered? (ii) what were their predicted consequences? (iii) why was τ^* preferred according to (7)? Vanilla agentic systems cannot answer any of these questions, because they do not generate or evaluate alternatives before acting.

B. Accountability

Because the selection criterion (7) is an explicit mathematical function of the consequence vector, accountability is formally traceable. If an executed strategy causes harm, an auditor can verify whether the harm was foreseeable given the historical consequence model (i.e., $\widehat{\mathrm{DD}}_{k^*}$) or resulted from an unforeseen regime change beyond the model's representational capacity. This distinction—between foreseeable and unforeseeable harms—is central to legal and regulatory accountability under the EU AI Act [7].

C. Harm Avoidance via Hard Risk Constraints

The drawdown constraint $\widehat{\mathrm{DD}}_k \le \rho_{\max}$ implements formal harm avoidance at the architectural level. Critically, this constraint operates on the *consequences* of proposed sequences—not their textual descriptions. Unlike guardrails applied to LLM output (which can be bypassed by adversarial prompts that describe a harmful strategy as benign), CAM evaluates strategies by *simulating* them. A strategy that describes itself as low-risk but exhibits large drawdown under simulation is filtered regardless of its description—a property uniquely resistant to prompt injection.

D. Human Oversight and Intervention

CA^{2}I's architecture naturally supports human-in-the-loop oversight. The consequence vectors $\{\phi(\tau_k)\}_{k=1}^{K}$ can be surfaced to a human supervisor before execution. The supervisor can override $\mathcal{S}$, choose a different τ_k from the evaluated set, or reject all candidates and trigger re-generation. This contrasts with vanilla agents, where the human must evaluate raw action sequences without any quantitative consequence estimate.

E. Generalisation to Other High-Stakes Domains

While instantiated in finance, CA^{2}I is domain-agnostic. In *clinical decision support* [3], CAM could simulate treatment pathways against electronic health records to estimate adverse event rates before a protocol is recommended. In *autonomous code execution*, CAM could run candidate code in a sandbox to detect sequences that consume excessive resources or modify protected files. In *supply chain management*, CAM could simulate logistics sequences against historical demand data to estimate stockout and overstock risks. In each case, the architectural pattern is identical: generate $\mathcal{T}$, assess $\{\phi(\tau_k)\}$, select τ^*, execute.

VIII. Conclusion

We presented Consequence-Aware Agentic AI (CA^{2}I), a formal framework that augments LLM agents with a pre-execution Consequence Assessment Module. CA^{2}I decouples generation from execution, inserting a principled simulation-and-selection step that evaluates predicted consequences of all candidate sequences before any irreversible action is taken.

Theoretically, we proved that risk-adjusted selection dominates random selection in expectation under stationarity (Theorem 1); quantified the dominance gap and showed it grows with the candidate-set size K (Proposition 1); and formalised maximum drawdown as a hard architectural risk constraint (equation (4)).

Empirically, on ten diversified US equities we found—reported candidly—that CA^{2}I attains a higher mean out-of-sample Sharpe (0.51) than uniform random selection (0.37) and than three established portfolio-optimisation selectors (mean-variance 0.19, risk-parity 0.45, minimum-variance -0.63), and reduces drawdown by 18% relative to buy-and-hold, while *not* increasing turnover and degrading gracefully under transaction costs. We also found, and disclose, that single-window pairwise differences among the active selectors are not statistically significant at $n{=}10$ (bootstrap CIs include zero), and that the *base* in-sample selector is statistically indistinguishable from random when pooled across a multi-regime rolling evaluation. This diagnosis motivated our enhanced consequence model, **CA^{2}I$^+$** (walk-forward cross-validation,

a robustness-penalised score, and a top-m ensemble), which *does* achieve a statistically significant, regime-robust edge: pooled over eight quarterly windows (n=80) it lifts mean out-of-sample Sharpe to 0.77 versus 0.53 for random ($+0.24$; Wilcoxon p=0.011; bootstrap CI $[+0.015, +0.45]$ excluding zero; positive in all 10 generation seeds) and beats the base selector by $+0.21$, while *reducing* drawdown. Performance also rises with K as the theory predicts, and Sharpe and Calmar selection agree on 100% of choices.

From a Responsible AI perspective—which we regard as the framework's principal contribution—CA^{2}I provides concrete, computationally tractable mechanisms for transparency, accountability, and harm avoidance, directly mandated by the EU AI Act, NIST AI RMF, and IEEE Ethically Aligned Design. Because the risk filter operates on *simulated consequences* rather than textual descriptions, it is inherently more resistant to prompt manipulation than purely textual guardrails.

Future work will extend CA^{2}I to multi-period adaptive consequence models, portfolio-level multi-asset selection, non-financial agentic domains including clinical decision support and autonomous code execution, and causal consequence models that reason about counterfactual outcomes beyond historical simulation.

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